



University of Engineering and Technology, Mardan

Week 4 Task Report

Student Survey – Python Data Analysis

Career Launchpad Phase II

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Date: 21/09/2026

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Week 4: Student Survey Data Analysis and Visualization (Python)

Introduction

Week 4 involved designing and analyzing a student survey covering demographics, habits, and preferences. Responses were analyzed in Python to uncover patterns in age, lifestyle, and study behavior.

Survey & Data Overview

- The survey collected responses on age group, coffee/tea preference, daily screen time, preferred study time, social media app usage, hobbies, number of languages spoken, and exercise routine.
- Responses were cleaned and grouped into categories before visualization.

Exploratory Data Analysis (EDA)

- **Age Distribution:** 19 students were aged 20–22 and 8 students were aged 23–25. (Figure 1)
- **Coffee or Tea Preference:** 63% of students preferred tea, 18.5% preferred coffee, and 18.5% preferred neither. (Figure 2)
- **Daily Screen Time:** 13 students reported more than 7 hours of screen time daily, 8 reported 4–7 hours, 5 reported 1–3 hours, and 1 reported less than 1 hour. (Figure 3)
- **Preferred Study Time:** Late night was the most popular study time (12 students), followed by morning (9), evening (4), and afternoon (2). (Figure 4)
- **Social Media App Preference:** WhatsApp was the most used app (16 students), followed by Instagram (7), Twitter (3), and Snapchat (1). (Figure 5)
- **Hobbies and Interests:** Traveling was the most common hobby (6 students), followed by Sports, Gaming, Reading, and Music (3 each). (Figure 6)
- **Languages Spoken:** Most students (18) spoke 3 languages, 7 spoke 2 languages, and 1 student each spoke 1 or 4 languages. (Figure 7)
- **Exercise Routine:** 63% of students reported not exercising at all, while the remainder exercised 1–6 times a week or daily. (Figure 8)
- **Age vs Study Time & Exercise:** Cross-tabulated charts showed that late-night studying and low exercise frequency were common across both age groups. (Figures 9–10)

Data Visualization

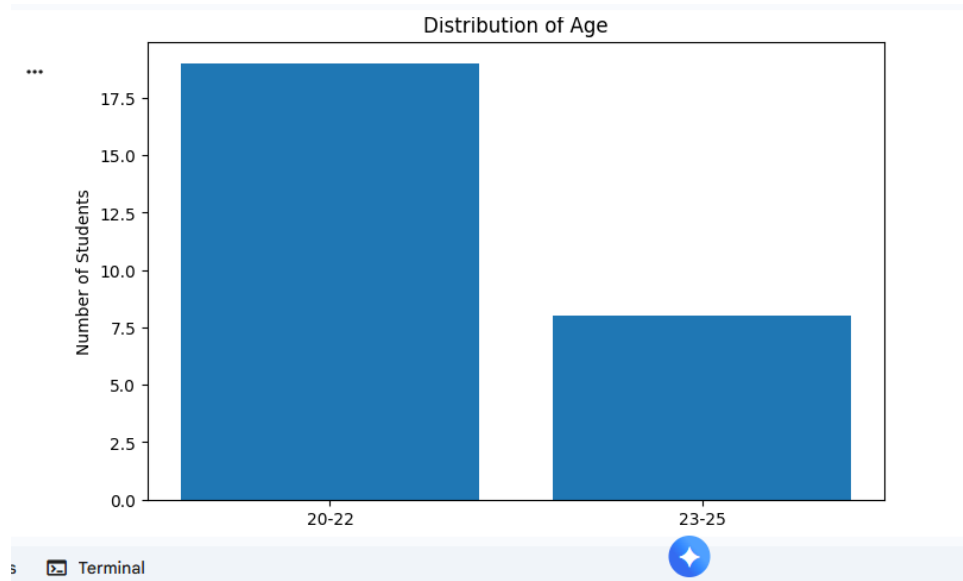


Figure 1: Distribution of Age

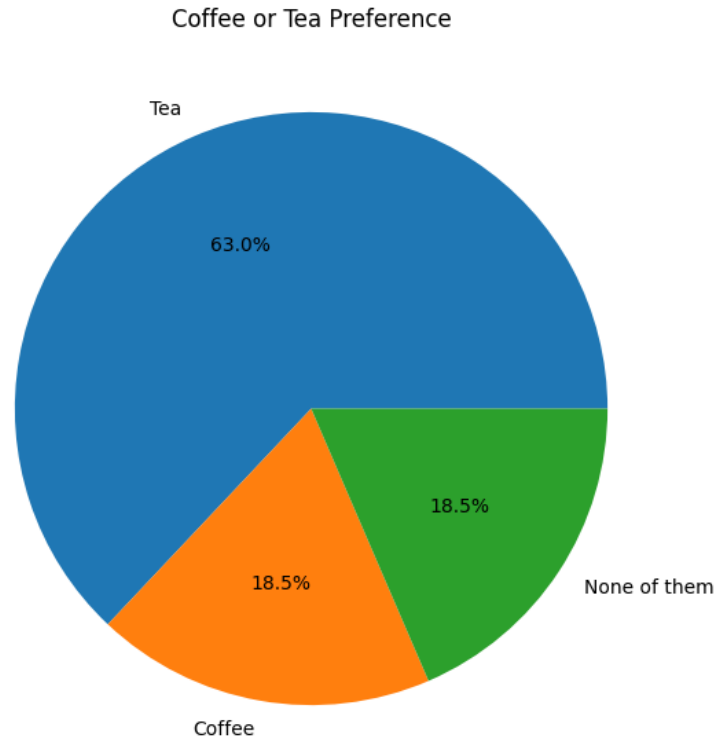


Figure 2: Coffee or Tea Preference

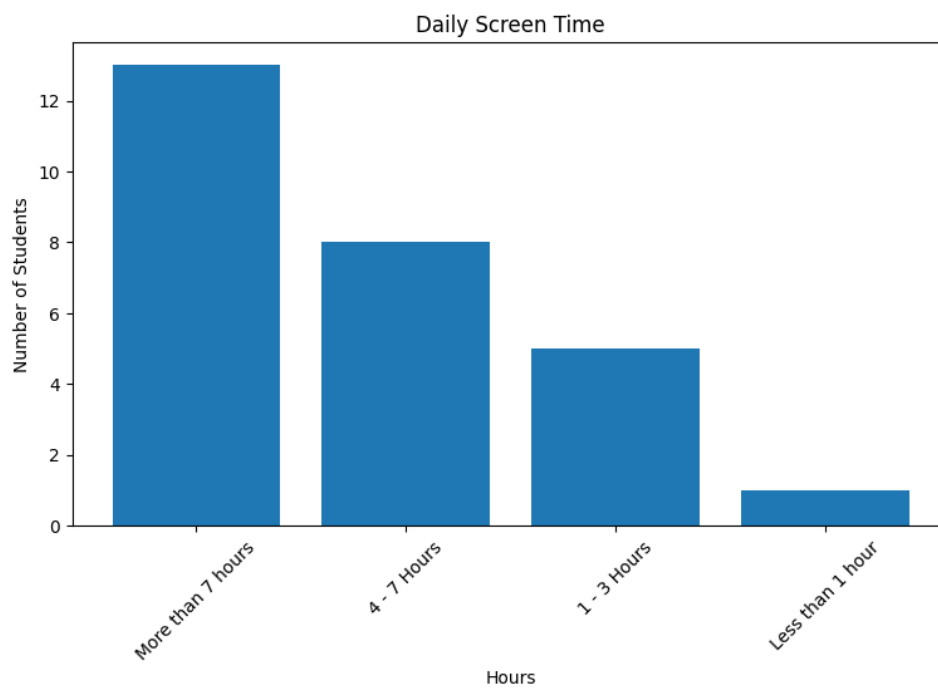


Figure 3: Daily Screen Time

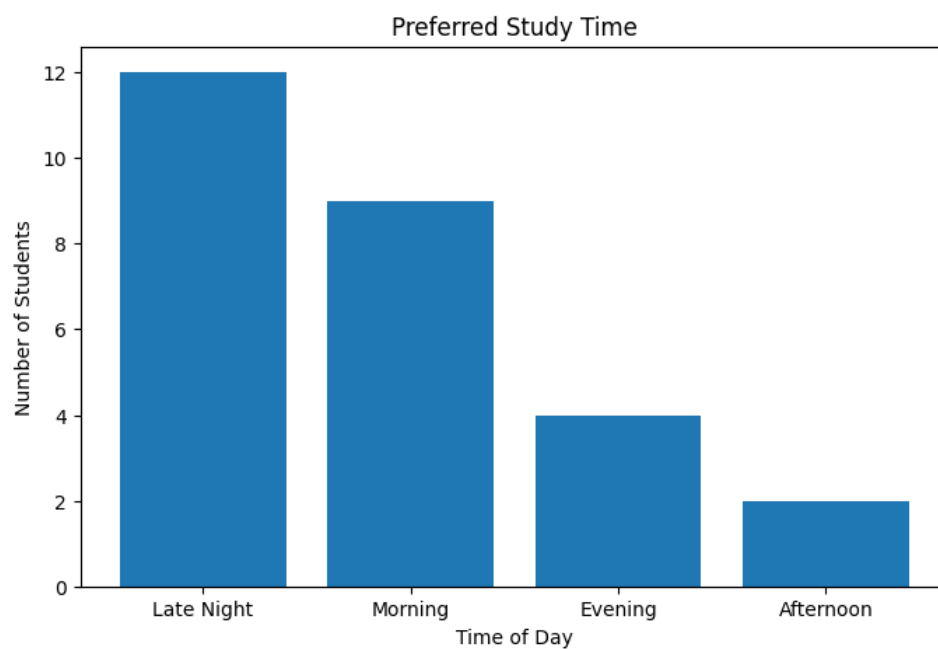


Figure 4: Preferred Study Time

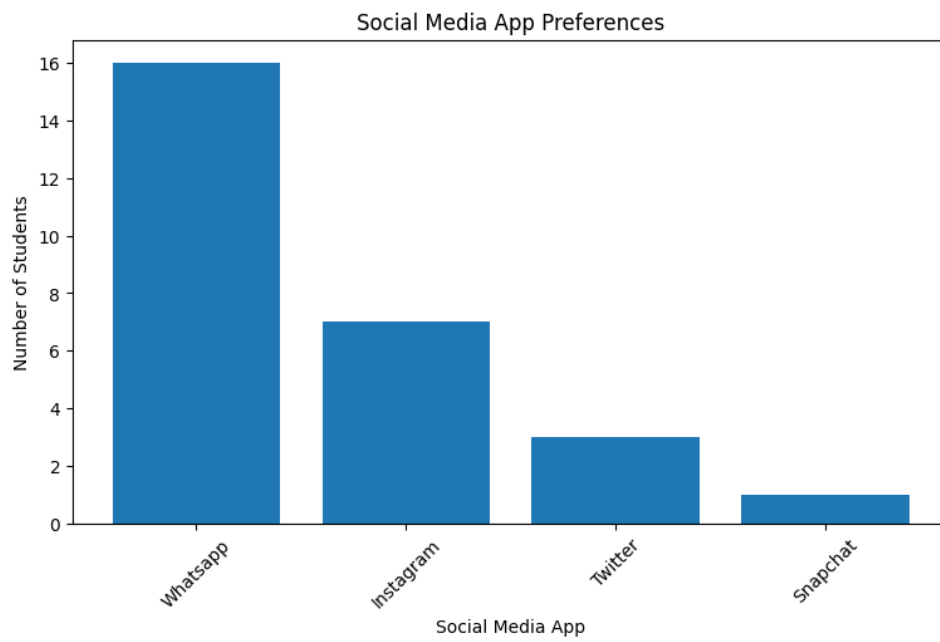


Figure 5: Social Media App Preferences

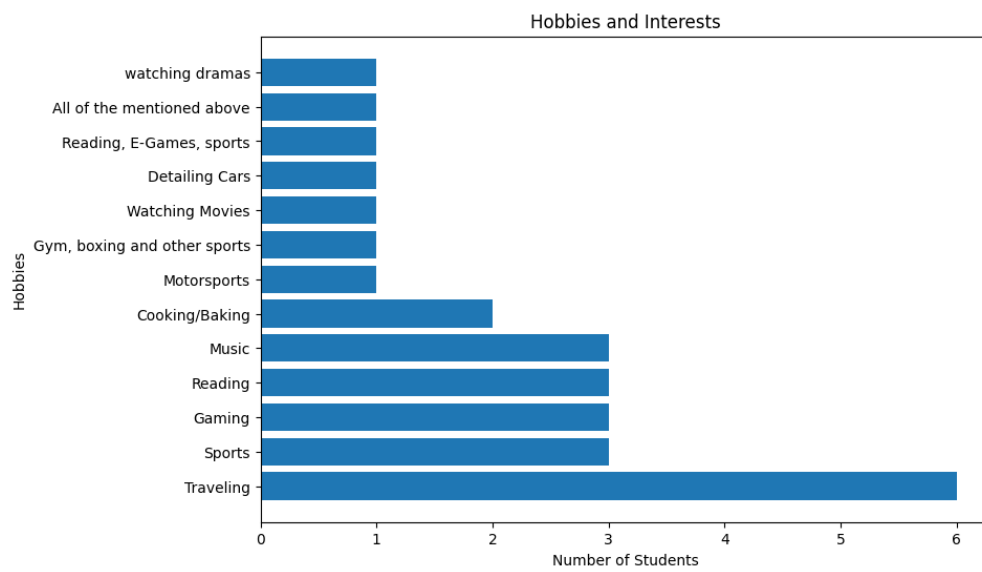


Figure 6: Hobbies and Interests

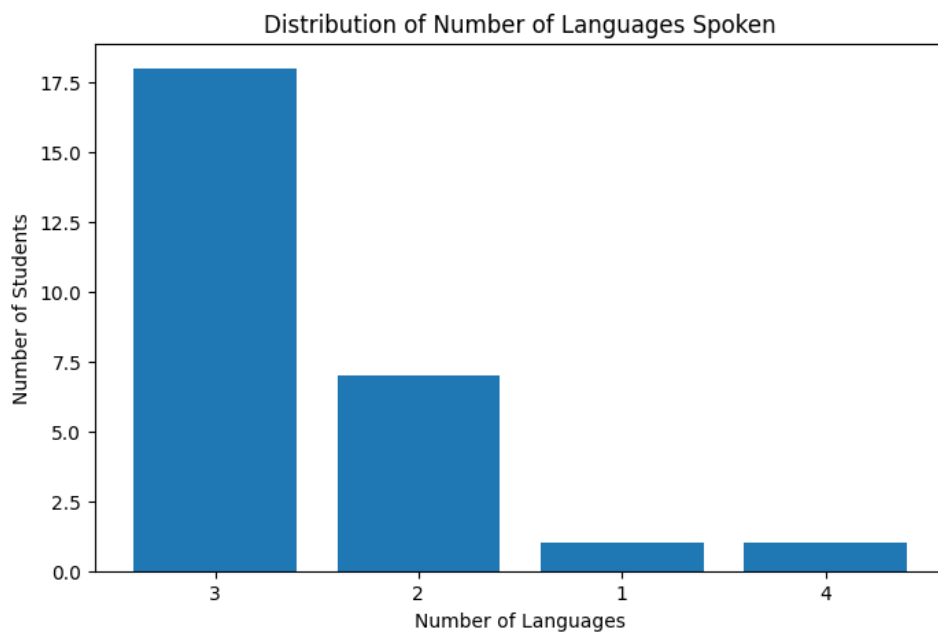


Figure 7: Distribution of Number of Languages Spoken

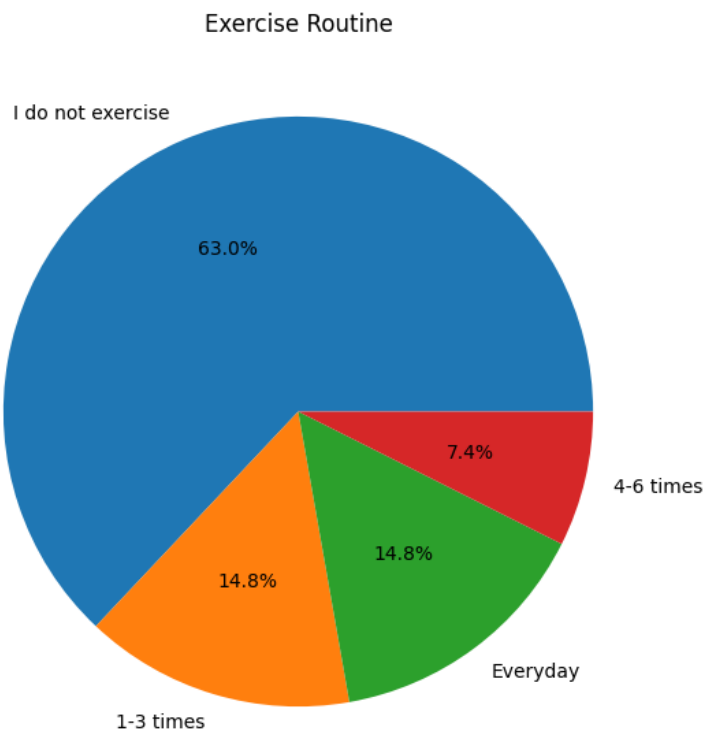


Figure 8: Exercise Routine

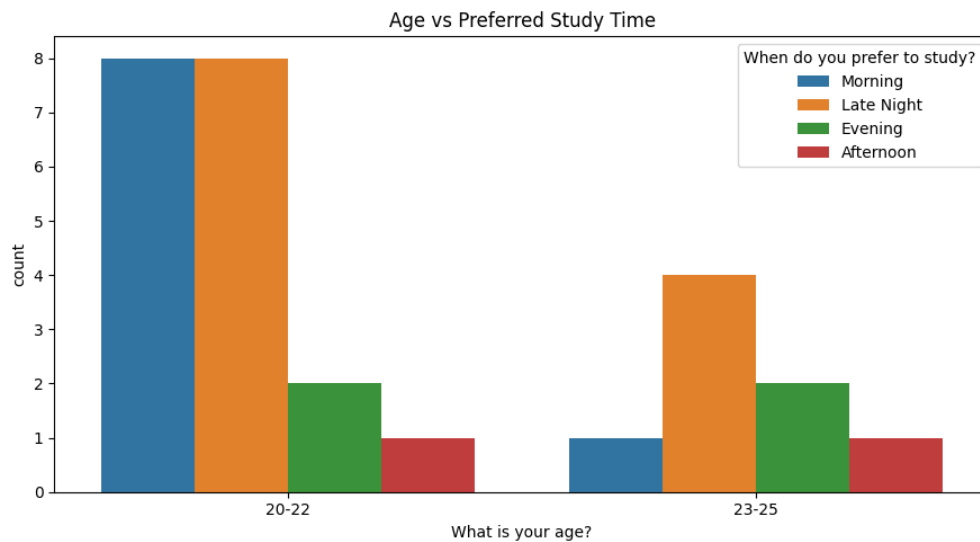


Figure 9: Age vs Preferred Study Time

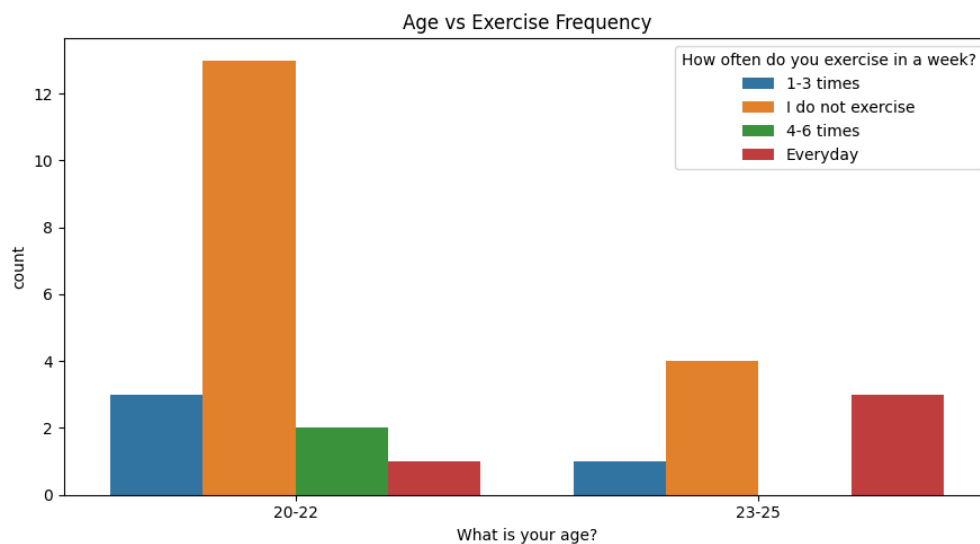


Figure 10: Age vs Exercise Frequency

Key Insights

- The majority of respondents were aged 20–22 and favored tea over coffee.
- Late-night studying and high screen time were common habits among students in both age groups.
- WhatsApp dominates as the preferred social media platform, and traveling is the most popular hobby.
- A large majority of students (63%) reported no regular exercise routine, indicating an area for wellness awareness.

Conclusion

Week 4's survey analysis in Python highlighted clear lifestyle and study patterns among students, particularly a preference for tea, late-night study sessions, and WhatsApp, along with a notable lack of regular exercise — insights that could inform future wellness or study-habit initiatives.